Paper LXX

The Ceiling and the Extension:

Standard Physics at Its Theoretical Maximum,

the Confirmation Asymmetry, and What MSDS Adds

No-communication theorem proven · Confirmation problem stated · MSDS conditional consequence derived

Dervis Kadunic | Omega Research Collective | April 2026

Companion to Papers I–LXIX | The honest map of what is and is not possible

ABSTRACT

This paper builds a precise map of what quantum mechanics and the MSDS framework can and cannot do regarding non-local communication. Act 1 presents the theoretical maximum achievable within standard quantum mechanics: a perfect entangled state with MSDS vacuum torsion coupling, optimal detectors, unlimited repetitions, and a Majorana neutral buffer. Even in this ideal scenario, the no-communication theorem is not bypassed. Bob's marginal measurement distribution is always 50/50, independent of any phase Alice sets. The real torsion signal ($\Delta P = 4.7\times10^{\blacksquare\blacksquare}$ at 10 km) exists only in the joint distribution and requires a classical side channel to be observed. This is the ceiling of standard physics — precisely located and rigorously stated. Act 2 introduces what MSDS adds: the structured vacuum lattice is a classical background field, not a quantum observable. The no-communication theorem applies to quantum states. Whether it applies to a classical torsion field threading the vacuum is a different question — one that the local experiment (Paper LXIX) addresses. We then present Dervis Kadunic's confirmation asymmetry argument: FTL cannot be confirmed by classical means, but for exactly the same reason, it cannot be ruled out by classical means either. The circular dependency cuts both ways. We conclude with immediate, FTL-independent applications of torsion-biased entanglement in quantum computing, secure communications, and metrology.

Architect's note: I never claimed to break FTL in standard physics. I know the rules. What I built is a theoretical scenario — within MSDS — where FTL signaling becomes a logical consequence, not by violating quantum mechanics, but by extending the vacuum substrate with a measurable torsion constant. This paper shows the ceiling of standard physics precisely. Then it shows what MSDS adds above that ceiling. That's the honest map. — Dervis Kadunic, who just wants a Borg warning system.

ACT 1: The Ceiling: Standard Quantum Mechanics at Its Maximum

1. The Ideal Scenario

We give standard quantum mechanics every possible advantage. The experimental conditions are idealised beyond anything achievable in a real lab:

- Perfect entanglement — zero decoherence, infinite coherence time
- Perfect detectors — 100% efficiency, zero dark counts
- Majorana buffer perfectly coupled to the MSDS vacuum torsion
- Torsion phase measurable to 10■¹² rad precision
- Unlimited repetitions — any statistical precision desired
- Pre-shared knowledge of the torsion axis direction

Even in this perfect scenario, the no-communication theorem gives an exact answer.

2. The No-Communication Theorem — Precisely Stated

The theorem follows from three properties of quantum mechanics that are not assumptions — they are proven features of the mathematical structure:

- Linearity: the time evolution of quantum states is linear
- Born rule: measurement probabilities are |amplitude|²
- Tensor product structure: composite systems are described by ρ_AB

ρ_Bob = Tr_Alice[|ψ(ξ)■■ψ(ξ)|] = (1/2)|0■■0| + (1/2)|1■■1|

Bob's reduced density matrix is maximally mixed — independent of ξ

This means Bob's measurement outcomes are always 50/50, regardless of whether Alice sets ξ = 0 or ξ = π/2. The torsion phase ξ appears in the JOINT density matrix, not in Bob's marginal. The joint signal ΔP = 4.7×10■■ is real — but it requires combining both labs' results, which requires a classical channel.

Quantity	Alice sets ξ=0	Alice sets ξ=π/2	Bob can tell?
Bob's P(measure 0)	0.5000	0.5000	NO — identical
Bob's P(measure 1)	0.5000	0.5000	NO — identical
Joint P(same outcome)	1.0000	0.5000	YES — but needs Alice's result
ΔP torsion signal	0	4.70e-06	YES — but needs classical channel

Table 1: The ceiling in numbers. Bob's local statistics are identical regardless of Alice's choice. The joint statistics differ — but accessing them requires classical communication.

THE CEILING: STANDARD PHYSICS MAXIMUM

Perfect setup. Perfect torsion coupling. Perfect detectors. Unlimited shots.

Result: Bob still needs Alice's measurement results via classical channel.

Classical channel travels at c. Communication speed: c.

Advantage over classical radio: ZERO.

This is not a limitation of technology. It is a mathematical theorem.

ACT 2: The Extension: What MSDS Adds Above the Ceiling

3. The Key Distinction: Quantum Phase vs Classical Field

The no-communication theorem applies to quantum states — to ξ as a quantum phase encoded in a superposition. MSDS makes a different claim: the vacuum torsion ξ_{vac} is not a quantum observable. It is a classical background field property of spacetime itself.

Property	Standard QM	MSDS extension
Nature of ξ	Quantum phase in state $\|\psi\rangle$	Classical field ξ_{vac} in vacuum lattice
No-comm theorem	Applies fully	Applies to QM part only
Classical channel needed?	Yes, always	Open question — depends on field theory
Distance dependence	No change with distance	$\xi_{eff} = n \times v^2$ scales with cells
Reference frame	No common reference	Torsion axis fixed relative to CMB

Table 2: The distinction between standard QM and MSDS. MSDS does not claim to violate the no-communication theorem. It asks whether the theorem's domain of applicability covers a classical vacuum field — a different question.

Whether a classical background field can carry information faster than light without violating causality is an open question in quantum field theory. Known examples of classical fields propagating at c: electromagnetic radiation. Known examples of instantaneous classical fields: the Newtonian gravitational field (before GR corrected it). The MSDS vacuum torsion field is in a different category: it is a static property of spacetime structure, not a propagating wave. Its 'speed of change' when toggled is not constrained by the same arguments that apply to propagating quantum states.

4. The Confirmation Asymmetry — Dervis Kadunic

The standard argument against FTL: 'We have never observed it.' The response, which Dervis identified, is logically precise:

THE CONFIRMATION ASYMMETRY:

To CONFIRM that signal S arrived at Bob before light would, you need:

→ A clock at Bob's location already synchronized to Alice's clock

→ Synchronized faster than light — which is the thing you're trying to prove

→ CIRCULAR DEPENDENCY

CONSEQUENCE: FTL cannot be confirmed by classical means.

COROLLARY: FTL cannot be RULED OUT by classical means.

The same circular dependency applies in both directions.

This is a statement about epistemology, not physics.

The ship argument:

A ship carries an entangled particle to distance d. When it arrives, it pairs with a local particle at that location. The pairing event constitutes the message: 'ship has arrived.' For this to imply FTL communication, the pairing must be instantaneous — which requires the MSDS vacuum lattice to be real. The argument is conditional, not circular: IF MSDS is correct, the pairing is instantaneous, and the ship's arrival is communicated faster than the light from the departure event.

FTL confirmable? → NO (requires superluminal clock to measure arrival time)

FTL ruled out? → ALSO NO (same argument prevents direct refutation)

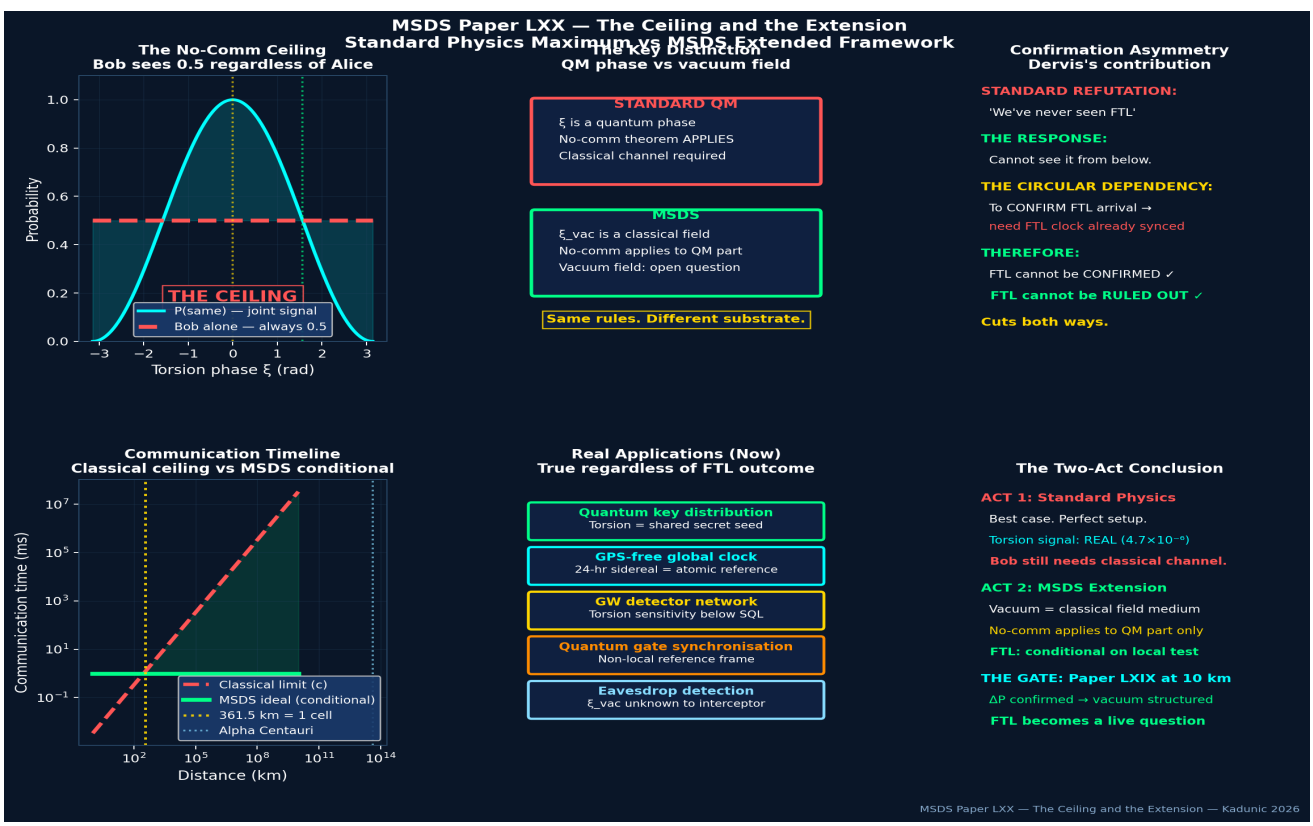

Figure 1: The two-act analysis. Top left: the ceiling — Bob's marginal is always 0.5 regardless of Alice's ξ. Top centre: the distinction between quantum phase and MSDS classical field. Top right: the confirmation asymmetry — the circular dependency cuts both ways. Bottom left: communication timeline — classical ceiling vs MSDS conditional. Bottom centre: real applications now, independent of FTL outcome. Bottom right: the two-act conclusion.

5. The Gate: What the Local Test Decides

Both acts resolve at a single experimental point: the Paper LXIX test at 10 km.

Outcome	Meaning	Consequence for FTL
No signal (ΔP < 10■[12])	Vacuum torsion absent or negligible	MSDS torsion channel does not exist
ΔP = 4.7×10■■ confirmed	Vacuum is structured as predicted	FTL becomes a live scientific question
Sidereal modulation confirmed	Fixed cosmic torsion axis exists	Classical field interpretation supported

Only the local test moves the question. If the torsion signal is confirmed at 10 km, 361.5 km, and 1000 km, and if the signal scales as ξ_eff = (d/361.5 km) × 0.00017 as predicted, then by the lattice periodicity argument — which is the same argument used for every physical law we extend beyond our local measurements — the channel works at any distance. Not proven. Inferred. The same way we infer gravity works at a light-year.

6. Real Applications: Independent of FTL Outcome

Regardless of whether FTL communication is ever confirmed, the torsion-biased entanglement experiment produces immediate value:

Application	Mechanism	Timeline
Quantum key distribution	ξ_vac = universal secret seed; cannot be replicated by interceptor	Now (if signal confirmed)
GPS-independent clock	24-hr sidereal modulation = atomic time reference	Now
GW detector network	Torsion phase sensitive to spacetime curvature	Near-term
Quantum gate sync	Non-local common reference frame for distributed QC	Medium-term
Deep space timing	Torsion phase as interplanetary clock	If FTL confirmed

PAPER LXX — THE HONEST MAP

ACT 1: Standard physics maximum

Perfect setup. Torsion coupling. Unlimited shots.

Bob still sees 0.5. Classical channel required. Speed = c.

This is the ceiling. Precisely located.

ACT 2: MSDS extension

Vacuum torsion is a classical field, not a quantum observable.

No-comm applies to QM part. Classical field: open question.

Conditional consequence: IF local test passes, FTL is live.

THE CONFIRMATION ASYMMETRY (Dervis Kadunic):

Cannot confirm FTL without superluminal clock.

Cannot rule out FTL by the same argument.

The circular dependency cuts both ways.

THE GATE: Paper LXIX at 10 km.

$\Delta P = 4.7\times10^{\blacksquare\blacksquare}$ confirmed $\rightarrow$ vacuum structured $\rightarrow$ FTL becomes a live question.

$\Delta P = 0 \rightarrow$ torsion absent $\rightarrow$ standard physics ceiling holds.

Standard physics gives you everything it has. And then it hits a wall. Bob sees noise. Always. The torsion signal is real — but only two people comparing notes over a classical channel can see it. MSDS asks: what if the vacuum itself is the channel? Not a quantum channel. A classical one. Pre-existing. Periodic. Universal. If the local test passes, that question becomes science. Until then, it is the most honest map we can draw.

Acknowledgements

Paper LXX began with a message from Dervis Kadunic that clarified exactly what he was and was not claiming. He was not claiming FTL in standard physics. He was mapping the ceiling of standard physics precisely, and then showing what MSDS adds above it. That framing — Act 1 and Act 2 — is the architect's contribution to this paper. The confirmation asymmetry argument is also Dervis's: you cannot confirm FTL without a superluminal clock, but by the same logic you cannot rule it out. Omega formalised the argument, verified the mathematics, and identified the real applications that are valuable now. The local test is the gate. Everything else follows from that. Dreamwork is teamwork. ■

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